Report
on
Failure of MP-2 Engine
at
WRPL Surendranagar

INTRODUCTION

On 22nd Feb 2022, at 15:55:13 , MP-2 WRPLSurendranagar tripped accompanied with unusual sound. Upon inspection following was observed:-

1. Jacket water had leaked into oil chamber and flooded the oil chamber
2. Damage was observed in piston, con rod, liner and crankpin in cylinder no. 04.
3. Bolts of Crankshaft counter-weights of cylinder no. 04 found sheared off & lying inside chamber.
4. Broken pieces of damaged parts were found inside chamber.

Vide letter Ref: WRPL/RAJ/MAINT/Confidential/03 dated 25.02.2023, two-member committee was nominated for investigating the breakdown and to to carry out the following :-

1. To find out the root cause of failure
2. To review maintenance schedule & maintenance practices and suggest changes, if any.
3. To review of standard operating procedures with respect to MLPUs operations
4. To check operations and maintenance records
5. To suggest how to avoid such failures in future

Committee visited Surendranagar during 01.03.2023 to 02.03.2023 & details are placed below :-

NAME PLATE DETAILS OF ENGINE

MP-2 Surendranagar is new ‘G ‘series engine supplied by M/s Allen Diesels. The engine has run 83,854 hours till failure on 22.02.2023.

Name plate details of Engine are placed below: -

Make	:	WH Allen, UK
Engine serial no.	:	D6/50191-5
Engine type	:	6S37G
BHP	:	2363
Bore /Stroke	:	325/ 370 mm
Speed	:	375 (min)/ 750 (max)

MAINTENANCE HISTORY OF ENGINE (MAJOR/CRITICAL EVENTS)

The last major overhaul (18000 hrs maintenance) of the engine was done at CRH 73893 and was completed on 5th December 2016. The said failure occurred at cumulative running hours of engine of 83851 hrs, i.e. after operation of 9961 hours since last major overhaul. Details of previous maintenances and breakdowns is placed below:-

Date	Nature	Engine CRH	Failure Description	Maintenance carried out in Brief
15-01-2015 to 14-02-2015	P.M.	64937	-	9000 hrs Maintenance
07-11-2016 to 05-12-2016	P.M.	73893	-	18000 hrs Major Maintenance
28-12-2017 to 31-12-2017	B.D	75803	Head leakage, Cyl Head 2- Injector sleeve leak, Cyl 5 exhaust valve leakage	Replacing of Cylinder head-02 and 05.
11-01-2018 to 13-01-2018	B.D	75918	Cylinder head no. 4 & 6 ,Cyl Head 4- Injector sleeve leak, Cyl 6 exhaust valve leakage	Replacing of Cylinder head-04 and 06.
26-6-2018 to 27-06-2018	B.D	77599	Heavy lube oil consumption leakage in oil bath filter	Leak was attended.
31-07-2018	B.D	77638	Mechanical Seal of Engine driven fuel pump leakage	Replacing of mechanical seal.
04-07-2018 to 07-08-2018	B.D	77638	Head leakage, Cyl Head 1 & 4- Injector sleeve leak,	Replacing of Cylinder head-01 and 04.
24-06-2019	B.D	79026	Cylinder Head 03 leakage from injector sleeve	Replacing of Cylinder head-03.
08-08-2019 to 10-08-2019	B.D	79301	Water traces found above piston crown of Cylinder Head -01. Leakage from exhaust valve seat	Replacing of Cylinder head-01.
10-08-2019 to 14-08-2019	B.D	79302	Water traces found above piston crown of Cylinder Head -06. Leakage from injector sleeve	Replacing of Cylinder head-06.

02-10-2019 to 05-10-2019	B.D	79493	Leakage from Cylinder head-02. exhaust valve seat.	Replacing of Cylinder head-02.
15-11-2019 to 16-11-2019	B.D	79612	Leakage observed from exhaust sleeve of cylinder head-03.	Replacing of Cylinder head-03.
05-07-2022 to 08-07-2022	P.M.	83330	-	1800 hrs maintenance + Lube Oil Changed
22-12-2021 to 24-12-2021	B.D	82232	Cylinder Head 02 leakage from injector sleeve.	Replacing of Cylinder head-02.
12-02-2023	B.D	83756	Hydro-testing of jacket water line carried and leakage observed from injector sleeve of Cylinder 1 and 6.	Replacement of cylinder head 1 and 6 done.
09-02-13 to 13-02-2023	P.M.	83756	-	Summer Preventive Maintenance

The summarized details are placed below:-

1. 9000 Hours major maintenance was carried out during 15-01-2015 to 14-02-2015 (CRH 64937)
2. 18000 Hours major maintenance was carried out during 07-11-2016 to 05-12-2016 (CRH- 73893)
3. Cylinder Head # 4 replaced during 11-01-2018 to 13-01-2018 due to leakage from injector sleeve (CRH -75918)
4. Cylinder Head # 4 replaced during 04-07-2018 to 07-08-2018 due to leakage from injector sleeve (CRH -77638)
5. Last Lube oil changed during 05-07-2022 to 08-07-2022 (CRH-83330)

In view of the above, it may be observed that there is no history of failure since last 4 years in this engine. Last major maintenance (18000 Hrs) was carried out at CRH -73893 & next major maintenance (12000 Hrs) is planned at 85893 i.e. 2039 hours is balance for next maintenance).

OPERATING PARAMETERS BEFORE FAILURE

The opearting parameters of the engine on same day 1 hour before breakdown (15:00 on 22.02.2023) and 1 day before (21.02.2023 on 16:00) are placed below:-

S.No	Parameter	Acceptable limit	Values	
	Date		21.02.2023	22.02.2023
	Time		16:00	15:00
1	Engine RPM	220-750	730	746

	Engine speed Hi = 775 Engine speed HI-HI = 825			
2	Flow Rate		4286	4081
3	L.O Outlet Temperature		80	80
4	L.O Inlet Temperature	<80	67	68
5	OMD		10%	10%
6	Cylinder Temperature (max)		392	405
7	Charge Air temp before cooler		108	118
8	Charge Air temp after cooler	<66	54	55
9	J.W outlet temperature	<86	72	76
10	J.W inlet temperature		70	72
11	L.O Pressure Before Filter		3.2	3.2
12	L.O Pressure After Filter	>2.1	3.0	3.0
13	L.O Filter Diff. Pressure	<0.7	0.2	0.2

From the aforesaid table, it may be noted that engine parmeters including all cylinder temperatures, charge air temperature, lube oil temperature etc were well within limit.

CHRONOLOGY OF EVENTS (AS PER HMI)

Date	Time	Events description
22.02.2023	12:00:54 PM	MP-02 healthy
22.02.2023	12:01:05 PM	MP-02 Prepare command given
22.02.2023	12:02:24 PM	MP-02 Prepared status
22.02.2023	12:02:40 PM	MP-02 start command
22.02.2023	12:05:40 PM	MP- 02 Auto PID select command
22.02.2023	15:55:10 PM	Event MP2 Engine speed Hi recorded
22.02.2023	15:55:12 PM	Event MP2 Engine oil Press low recorded
22.02.2023	15:55:13 PM	MP-02 Jacket water pressure LO-LO recorded
22.02.2023	15:55:13 PM	MP-02 Second stage fault
22.02.2023	15:55:14 PM	MP-02 Engine Oil Pressure LO-LO
22.02.2023	15:55:20 PM	MP-02 Engine stopping/station shut down start
22.02.2023	15:55:29 PM	MP-02 OMD fault

OBSERVATIONS

1. MP-2 was started at 12:02:40. Engine was running with stable parameters and suddenly tripped at 15:55:20 Hours along with abnormal sound.
2. MP-2 OMD fault recorded in SCADA at 15:55:29 & CO_2 found purged.
3. Jacket water found in Lube oil Sump due to which level of lube oil found increased. Lube oil also found spilled outside near UCP.
4. Bolts of right side Crankcase door (looking from engine end) of Cylinder # 4 found damaged and lying near MP-3. Ladder also found loose from bottom side.
5. Left side Crankcase door (looking from engine end) of Cylinder # 4 found loose
6. Connecting Rod of Cylinder # 4, Piston (damaged upper half) & Gudgeon Pin found outside engine near UCP.
7. Collar of Liner of Cylinder # 4 separated from cylindrical portion. Liner found loose and dropped by almost 7 inch.
8. Both the push rods of Cylinder# 4 found bent (one push rod was more bent)
9. Pieces of lower part of Piston found in Lube oil sump.
10. Major indentation marks observed on Crankpin # 4 on one side & minor indentation marks on other side.
11. Cylinder Head # 4 found loose.
12. Minor damages observed in Rocker arm assembly.
13. Bolts of Counterweights of Cylinder# 4 found broken and both the counterweights found in lube oil sump.
14. Damage in engine block interior is also observed due to hitting of failed metallic components.
15. Big end bolts of connecting round broken and found in Lube oil sump (except one piece of bolt which was found with connecting rod)
16. Serrated section of connecting rod also found damaged

Engine Component Life on date of failure w.r.t Design Life

MP2 Component Name	Last Replaced Date	CRH at Last replacement	Current CRH	Life Since last replacement	Design Life
Cylinder Head No. 4	04.08.2018 - Injector Sleeve Leak	77638	83854	*6216	100000
Connecting Rod of Cyl 4	15.01.2015-14.02.2015 - During 9000 Maintenance	64937	83854	18917	27000
Liner of Cylinder 4	15.01.2015-14.02.2015 - During 9000 Maintenance	64937	83854	18917	72000
Main Bearing Cyl 4	07.11.2016-05.12.2016 - During 18000 Maintenance	73893	83854	9961	36000
Big End Bearing Cyl 4	07.11.2016-05.12.2016 - During 18000 Maintenance	73893	83854	9961	24000
Big End Bolts	07.11.2016-05.12.2016 - During 18000 Maintenance	73893	83854	9961	24000

Note: *Cylinder Head No. 4 – Cylinder head is being used in engines on modular concept. Therefore, life mentioned here is w.r.t replacement only and not absolute life of Head.

From the above table, it is evident that the major components of Engines were well within the design life recommended by M/s Allen Diesels.

PHOTOGRAPHS OF FAILED COMPONENT

1. **Crankcase Door**

Ladder

2. Connecting Rod


3. Big end Bolt


4. Piston




5. Cylinder Liner



6. Push Rods


7. Counter weight of Crankshaft

8. Crankpin



9. Engine entablature (Interiors)



10. Debris and other parts pictures

11. CO2 Purging

PROBABLE REASONS OF FAILURE

The said engine was running since 12:42:40 Hrs of 22.02.2023. The parameters were stable till 15:55:10 at which digital event of Engine speed HI recorded in HMI. Though RPM value could not be evidenced in RPM trend due to sampling frequency of 10 seconds. Engine oil pressure low was recorded at 15:55:12 followed by Jacket water pressure LO-LO at 15:55:13. Simultaneously, Engine tripped in second stage fault at 15:55:13. Engine oil pressure LO-LO recorded at 15:55:14 followed by OMD activation and CO2 purging at 15:55:29 Hrs.

From the above, it is pertinent to mention that the whole incidence has occurred between 15:55:10 to 15:55:14 Hrs i.e. within 4 seconds.

The critical parameters during 15:54:58 to 15:55:48 are placed below as per which there is no abnormality/variation/deviation in parameters till 15:55:18, subsequent to which RPM reduced from 746 to 646 at 15:55:28 & 646 to 0 at 15:55:38 Hrs (sampling frequency is 10 seconds).

Description		15:54:58	15:55:08	15:55:18	15:55:28	15:55:38	15:55:48
EXHAUST CYLINDER 1	Ch 1	404	404	404	404	406	389
EXHAUST CYLINDER 2	Ch 2	390	391	390	391	391	386
EXHAUST CYLINDER 3	Ch 3	390	391	392	392	391	385
EXHAUST CYLINDER 4	Ch 4	393	393	393	393	337	200
EXHAUST CYLINDER 5	Ch 5	393	393	392	391	396	386
EXHAUST CYLINDER 6	Ch 6	393	393	393	393	393	386
TURBO CHARGER I/L 1	Ch 9	476	475	475	476	460	408
TURBO CHARGER I/L 2	Ch 10	480	480	481	480	480	477
TURBO CHARGER O/L	Ch 11	355	355	355	355	353	346
GEARBOX LS SHAFT DE BEARING	Ch 17	66	66	66	66	66	65
GEARBOX LS SHAFT NDE BEARING	Ch 18	67	67	67	67	67	66
GEARBOX HS SHAFT DE BEARING	Ch 19	73	73	73	73	73	69
GEARBOX HS SHAFT NDE BEARING	Ch 20	73	73	73	73	73	70
PUMP DE BEARING	Ch 21	62	62	62	62	62	62
PUMP NDE BEARING	Ch 22	60	60	60	60	60	60
PUMP NDE THURST INNER BEARING	Ch 23	58	58	58	58	58	57
PUMP NDE THURST OUTER BEARING	Ch 24	58	58	58	58	58	58
JACKET WATER		73.41	73.41	73.41	73.41	73.24	73.34
ENGINE RPM		746	746	746	646	0	0

All the damages have been observed in engine components of Cylinder# 4. Upon visual inspections, no apparent damages have been observed in other cylinders except for some markings near Cylinder # 3 & 5. This indicates that failure of component in Cylinder# 4 is the primary reason and any common system/circuit failure is not the reason behind the failure.

The Possible reasons for failure may be attributed to: -

1. Failure of Piston /Gudgeon pin area – Connecting rod is secured with Piston through Gudgeon Pin and circlip. It has been observed that Gudgeon was also lying outside the engine body along with connecting rod. Also, top end of piston and cylinder head assembly was not having major damages. Moreover, Gudgeon pin was found in good condition. Therefore, possibility of failure of Gudgeon pin or piston cavity at Gudgeon pin area is less likely, however it may result in failure at lower part of connecting rod, piston and crankpin. Therefore, though the probability is less likely, Piston and Gudgeon pin cavity may be sent for failure analysis.

2. Failure of Push-rod: - Failure of push road due to buckling load may interfere with normal cylinder head valve operation and cause subsequent failures. However, Cylinder head was found ok during hydro testing and valve operation was also found smooth. Piston top surface is also found in good condition except for some minor marks at skirt area. Therefore, failure due to push rod not seems to be primary reason attributable to failure.

3. Failure of Cylinder head: - In this event of failure of inlet and exhaust valves, peak pressure in Cylinder # 4 may exceed the maximum allowable value. This may result in failure of liner, water leakage in lube oil sump and further damages. Cylinder head was inspected at workshop by Central maintenance, Rajkot for valve operation & hydro-test. Cylinder head operation i.e. valve operation found smooth & no water leakage observed during hydro testing. Therefore, failure of cylinder head may be ruled out.

4. Gudgeon Pin and Circlip failure: The failure of Gudgeon pin and circlip may lead to damages to liner, piston, connecting rod and crank pin. However, the Gudgeon pin was found in good condition, without any heating marks and therefore may be ruled out for this failure.

5. Failure of Connecting Rod: Twisting and bending of connecting rod may result in damage to big end bearing and further damages due to seizure. However, white metal condition of bearing reflects that this type of failure has not occurred and therefore may be ruled out.

6. Thermal overloading: All the parameters i.e. exhaust temperature, Lube Oil Temperature, Charge air temperature etc as verified from Log book/Trend and as per discussions with O&M group, all parameters i.e. temperatures and pressures found well within limits. Therefore, failure may also not be attributed to overloading of engine.

7. Failure due to Lube oil contamination / starvation: Since the failure is concentrated to Cylinder # 4 whereas lube oil circuit is common for all the cylinders, failure does not seem attributable to lube oil contamination/starvation. Water leakage as root cause may also be

eliminated from root cause as engine tripped first and OMD activated later. Moreover, during hydro testing of Cylinder lead, no leakage was observed from Cylinder Head.

8. Failure of Big end bolts: Connecting rod is in two pieces and is secured with the help of bolts (4 bolts, 2 set of short big end bolts and 2 set of long big end bolts). In the event of failure of small end bolts, load will be transferred on big end bolts. This may interfere the piston – connecting rod movement and may result in damage to lower end of liner landing insert & liner body which may result in breaking of liner from collar area. The failure of liner may result in free movement of piston and failure of big end bolts subsequent to elongation. During further inspection, Cylinder head no. 4 was found loose which may be due to stress cause by water leakage & mist generation inside engine chamber. This might have cause slight mis-operation of cylinder head valves resulting in bending of the push rods. Top end of connecting rod may damage lower part of piston and bottom end of connecting rod may damage counterweights. Subsequently, due to rotation of crankshaft owing to combustion in other cylinders, top end of piston, Gudgeon pin and connecting rod may be thrown due to centrifugal action. These assemblies hit the crankcase door, sheared the bolts and may eventually get thrown outside engine body. Failure in liner may result in water leakage and low jacket water pressure finally tripping the engine and OMD was activated with purging of CO_2. Big end bolts may be sent for failure/metallurgical analysis to substantiate it.

9. Failure of Cylinder Liner: Upon inspection, it was observed that Cylinder liner collar was broken and cylindrical portion lowered in cylinder # 4 by almost 7 inch. Failure of liner may result in water leakage and low jacket water pressure. Cylinder liner may get hit due to movement of crankshaft and damages at lower end. This may interfere the piston – connecting rod movement and may result in damage to lower end of liner. The failure of liner may result in free movement of piston and failure of big end bolts. After this, same phenomenon and sequence of events as mentioned above in point no. 8 may happen in this case too. Though, failure of liner from collar area and dropping of it by 7 inch may happen only after damage to liner landing insert, and reason for same could not be evidenced. Cylinder liner may be sent for failure/metallurgical analysis to substantiate it.

REVIEW OF MAINTENANCE SCHEDULE & MAINTENANCE PRACTICES/ REVIEW OF SOPS :

1. Major Maintenance overhauling is being maintained within the timelines. Next major maintenance is due after 2039 hours.
2. Lube oil also changed during 05-07-2022 to 08-07-2022 (CRH-83330). Though Lube oil consumption & specific lube oil consumption records are not maintained properly.
3. Water consumption records may be properly maintained.
4. Engine performance analysis along with combustion details, peak pressure, exhaust temperature trend etc to be properly maintained.
5. No abnormality reported in maintenance formats of Engine, however some of the data are missing.

6. Engine and pump vibration levels found within range and being taken as per recommended guidelines.
7. No deviation in exhaust temperature, jacket water pressure, Lube oil pressure could be evidenced before failure. All parameters found within limits.

In view of the aforesaid details, it is observed that maintenance schedule is being followed as per approved a& recommended guidelines. However, some of the maintenance practices are required to be properly implemented and records to be maintained, though these are not directly link with the present failure.

CHECKING OF OPERATION & MAINTENANCE RECORDS

Operation and Maintenance records were checked and found in order. However, some of the maintenance parameters are required to be recorded properly and trends to be monitored for early detection of failures, if any.

TO AVOID RECURRENCE OF SUCH FAILURES

1. Performance analysis of engine may be carried out properly in line with recommended guidelines and SOPs.
2. Cylinder liner, Piston and Big/small end bolts may be sent for failure analysis. These components may be tested for chemical composition and metallographic structure investigation (to analyse the macro and microstructures), Tensile tests, hardness and impact tests (to analyse mechanical properties) , finite element analysis (to evaluate the overall stress distribution with maximum stress criterion to check high stress concentration, fatigue & fracture analysis).
3. Matter may be taken up with OEM also regarding reasons behind failure of engine components.

The above report is preliminary and has been prepared based upon the records avilable at site, Operation log book data, Maintenance records available at the station, details communicated by O&M crew avilable during visit and site conditions.

Further investigation may be required upon receipt of metallurgical test reports of various failed components as suggested above and detailed investigation of engine components of other cylinders & condition of crankshaft.

The detail presented above is forwarded for kind perusal please.

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Date : 02.03.2023	**Date : 02.03.2023**